Introduction to Data Centre Management Using a Simulation Game

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Abstract. The abstract should briefly summarize the contents of the paper in 150–250 words.

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1 Introduction

The video game industry has experienced substantial growth over the past decade. In many countries, its economic impact has already surpassed that of traditional media sectors such as film, television, and newspapers. As the industry continues to expand in both scale and relevance, the application of video games increasingly extends beyond entertainment.

Serious games and simulation-based games provide players with interactive experiences that incorporate educational content. However, studies such as *Game-Based Learning with Computers: Learning, Simulations* [8] demonstrate that even games developed exclusively for entertainment require players to acquire new knowledge, develop complex skills, and devise effective strategies in order to progress through levels and overcome challenges.

Similarly, research such as *Video Games and Indirect Learning* by Monica Miller [7] suggests that the content of video games can also facilitate indirect learning, that is, learning that occurs without the game being explicitly designed for educational purposes.

Motivated by these findings, this work aims to investigate whether players acquire knowledge while interacting with **Port 0**, a resource management video game in which the player is responsible for operating a small data center within a fictional setting. Throughout the game, players must purchase, configure, and manage servers, fulfil client requests, and defend the infrastructure against cyberattacks.

Although the game is set in a science fiction universe, it incorporates several simulation-based mechanics, including word-based mini-games and server configuration tasks. The objective is to provide an engaging and enjoyable gameplay experience while simultaneously promoting knowledge acquisition, both through direct learning mechanisms embedded in the mini-games and through indirect learning arising from the contextual environment in which the player operates.

2 Related Work

The number of video games that convey knowledge to players, either directly or indirectly, is considerably large, regardless of whether they fall within the category of serious games. Alongside the video game industry, several authors have conducted studies and research on learning through video games in recent years. Trends related to the technology field, such as **data center** and **technology**, have also become increasingly popular in recent years.

Therefore, throughout this section, the objective is to analyze the learning strategies applied to video games and understand how they produce learning outcomes. Finally, the growth of trends related to the context of the game (**Port 0**) is analyzed.

2.1 Learning through observation

Learning through observation, also referred to as observational learning, is the process by which beginners absorb rules, initial steps, and complex procedures simply by watching experienced players or interacting with simulations and/or tutorials [8]. When integrated into video games or simulation systems, this type of learning acts as a pedagogical "scaffold" for the learner [8].

Several titles, such as CodeCombat, Manufactoria, and Labor Support Game, employ this learning strategy.

In CodeCombat, for example, the developers implemented a split-screen interface where the code is highlighted line by line while the hero performs the corresponding actions on screen, allowing players to directly observe the outcome of each command [9].

A similar approach to CodeCombat was adopted to convey the meaning of commands within the WordRush minigame. This minigame is discussed in Subsection 3.2.

2.2 Self-tutorage and Trial and Error

Learning through trial and error is an active and exploratory learning strategy in which individuals test hypotheses in real time to determine which actions lead to successful outcomes and which result in failure [9].

Trial and error is also a process in which learners repeatedly attempt missions or tasks until they achieve success or the highest possible score, allowing them to understand complex concepts through direct experience with their own mistakes [10,11].

Feedback loops can be used as a mechanism to enhance this technique, as they create a repetitive cycle in which the system responds to the player's actions, thereby providing continuous learning opportunities [12].

Accordingly, the developed project, **Port 0**, incorporates gameplay loops in which the player is required to repeat a set of tasks. This aspect is discussed in greater detail in Section 3.

Finally, self-tutorage is a common practice, particularly in fields such as programming, involving the development of solutions through continuous testing of what works and what does not in real time. This process functions as a form of self-learning and continuous skill development [13].

2.3 Indirect learning

Indirect learning is a process in which knowledge is acquired as a by-product of another activity, such as leisure [14,13]. This process occurs implicitly, unconsciously, or naturally, often without the individual initially realizing that learning is taking place [14].

This learning mechanism is effective because, rather than explicitly teaching content, games create environments in which knowledge is constructed through experience, problem-solving, and interaction with complex systems [7].

Commercial games such as *No Man's Sky* [15] and *The Legend of Zelda: Breath of the Wild* [16] nevertheless promote knowledge and skills such as:

- Experimentation;
- Scientific thinking;
- Creativity;
- Complex problem-solving;
- Resource management.

According to the study conducted by Monica Miller, *No Man's Sky* [15] fosters scientific curiosity not because it explicitly teaches science, but because its gameplay structure encourages players to adopt behaviors similar to those used in scientific inquiry [7].

Similarly, **Port 0** seeks to promote scientific curiosity through its environment, visual elements, and gameplay tasks.

2.4 *Hype* surrounding the technology field

Hype is defined as "a situation in which something is advertised and discussed in newspapers, on television, etc. a lot in order to attract everyone's interest" [17].

To evaluate the hype surrounding topics such as *technology* and *data center*, the *Google Trends* tool was used. Google Trends measures the relative search

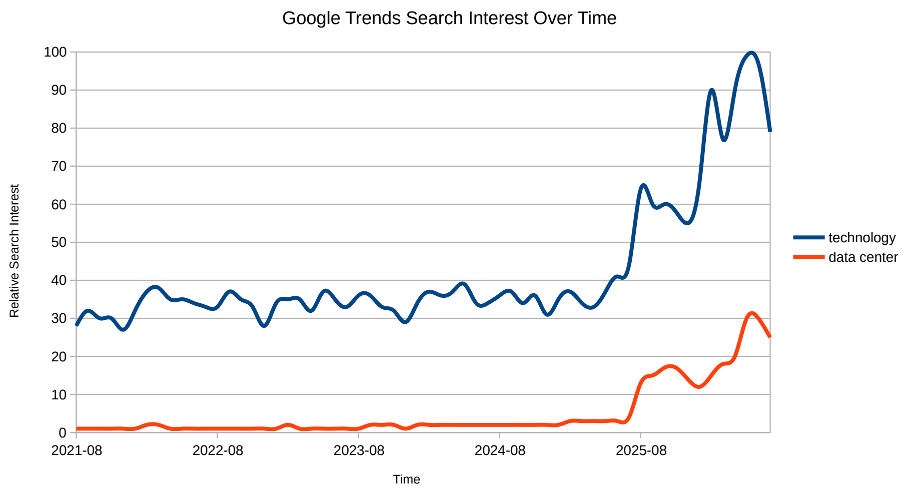

Fig. 1. Relative Search Interest for *Data Center* and *Technology*.

interest for terms entered by users into Google's search engine over time and across different geographical regions.

The data obtained from *Google Trends* are presented in Figure 1, which shows the worldwide relative search interest over time for the topics *data center* and *technology*.

From these data, it can be concluded that both topics have experienced substantial growth over the past five years. Although it may not seem obvious, there is a direct relationship between public interest and the motivation to learn.

Public interest in a particular topic or product may be the result of marketing efforts. When applied to a game and combined with an engaging context, it can increase the player's level of interest even before gameplay begins [8].

Furthermore, the technological domain in which the project is situated is itself a motivating factor for learning. Young people aspiring to careers in technology, such as game development, are often attracted to computer science programs because they revolve around an "object of pleasure" (the game) [13].

Therefore, the developed project, ***Port 0***, has strong learning potential, both direct and indirect, since it is situated within a field experiencing increasing hype. The attractiveness of an educational game is closely related to its content and to the connection that learners establish with its subject matter [10].

3 Design and development of XX game

3.1 Design of XX game

O conceito do jogo é descrito da seguinte forma: Um cenário fantasios com elementos de fição ciêntifica onde o jogador terá de intrepetar o papel de um engenheiro dentro um ambiente que remete a um data centers, onde ele terá que realizar tarefas tecnicas. Este tarefas seram apresentadas ao jogador sobre a forma de minigame de palavra, onde o mesmo terá que escrever um conjunto de comandos com uma dada finalidade. Além disso o jogador terá que se defender de ataques utilizando para isso de um canhão.

O jogo também apresenta vários loops de ações, entre os quais se destacam o loop de tarefas e o loop de ataques. O loop de missões é o mais comun e o que dura mais, é nele que o jogador recebe pedidos de clientes, onde terá de escolher entre aceita-los ou rejeitálos. Em caso de aceitação terá de resolver um minigame palavra inspirado na escrita de comandos em um terminal, onde o mesmo terá de escrever um conjunto de comandos para completar a tarefa. Já no loop de ataques, o jogador terá de usar um canhão para se defender de hordas de inimigos que viram atacar o espaço do data center.

Posto isto, várias tecnicas de design foram aplicadas no sentido de atingir o conceito idealizada da melhor forma possivel, tais como:

- Framework MDA [18];
- Flow [19];
- Hook model [20].

O framework MDA [18] foi utiliza como guia e construção do jogo, premitindo defenir as Mechanics, Dynamics e Aesthetic do projecto.

O flow [19], foi uma das tecnicas mais importantes, pois ela ajuda a medir e a projectar uma experiência que prenda o jogador em um estado flow, isto prenda um jogador em um estado de equilibrio, onde o mesmo não se encontre em um estado de boredom nem de anxiety.

O loop de ataques, foi pensando justamente para manter o jogador em estado de flow, uma vez este tipo de loop quebra a gameplay monótona das tarefas e estimula o jogador, fazendo assim o jogador se afastar do estado boredom.

Por fim, o Hook model [20], surge como uma estrutura teórica e prática desenhada para fomentar a criação de hábitos através da interação contínua com o produto. Além disso o modelo ajuda a defenir hábitos através dos quais é possivel criar interações contínuas com o produto, mesmo caso o jogo desenvolvido.

3.2 Development of XX game

Port 0, foi desenvolvido utilizando um conjunto de ferramentas e de tecnologias, tais como: *Unity* como engine grafica, *C#* como linguagem de programação, *Blender* como software de criação de modelos 3D e de texturização dos mesmos.

O jogo é composto por várias sistema que se relacionam com o objetivo de criar uma experiência completa e interativa. O sistema da loja, premite ao jogador comprar um servidor, no entanto para isso o jogador terá que passar por um conjunto de etapas onde terá de configurar vários componentes. Com isto a ideia seria introduzir os diferentes componentes que compõem um servidor ao jogador, além de fornecer uma pequena descrição à cerca de cada um.

Juntamente com o sistema da loja foi desenvolvido um sistema de economia, que é responsável por gerir as transições realizadas pelo jogador, como compra e venda. Além disso controla as transações periódicas, como pagamentos sistemáticos realizados por clientes devivo a serviços e pagamento de despesas como energia.

O sistema de tarefas é responsável por escolher tarefas randomicamente e as apresentar para o jogador. Este sistema é o que possibilita ao jogador ganhar dinheiro, quer de forma ativa realizando novas tarefas quer de forma passiva com pagamentos oriundos de tarefas prévias. Este sistema é um loop continuo que atua durante toda a duração da gameplay. O minigame (*WordRush*), onde o jogador escrever conjunto de comandos de Linux, é uma consequência deste sistema, uma vez que a resulução deste minigame é o que encerra a tarefa aceite.

Por fim o sistema de combate, é o ultimo sistema, e é responsável por gerir os ataques harckers que acontecem. Ele funciona de forma periódica e a sua função é quebrar o cliclo de gameplay das tarefas, atuando como elemento de rotura que interrompe uma gameplay que pode ficar monotona, dando um pouco mais de ação e divertimento para o jogador.

4 Results

5 Conclusion

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